Example 1.5 (Simple N*-Body Molecular System)* We can now describe a molecular system consisting of N neutral atoms with pairwise Lennard-Jones potentials; this could be used to model liquid argon or a similar inert system; it is one of the first important simulations performed using a smooth classical potential [306, 387]. The Lennard-Jones potential is

$$\varphi_{\mathrm{LJ}}(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right],$$

with typical parameters (for argon) $\epsilon = 1.65 \times 10^{-21}$ J and $\sigma = 3.4 \times 10^{-10}$ m. The energy of the system is

$$E = \frac{1}{2}\sum_{i=1}^{\mathrm{N}} m\dot{\boldsymbol{q}}_i^2 + \sum_{i=1}^{\mathrm{N}-1}\sum_{j=i+1}^{\mathrm{N}} \varphi_{\mathrm{LJ}}(r_{ij}),$$

where $r_{ij} = \|\boldsymbol{q}_i - \boldsymbol{q}_j\|$ and the mass m of an argon atom is 6.69×10^{-26} kg. The equations of motion are, for $i = 1, 2, \ldots, \mathrm{N}$, using the chain rule,

$$\begin{aligned} m\ddot{\boldsymbol{q}}_i &= \sum_{\substack{j=1\\ j\neq i}}^{\mathrm{N}} \frac{\varphi'_{\mathrm{LJ}}(r_{ij})}{r_{ij}}(\boldsymbol{q}_i - \boldsymbol{q}_j) \\ &= -24\frac{\epsilon}{\sigma}\sum_{\substack{j=1\\ j\neq i}}^{\mathrm{N}} r_{ij}^{-1}\left[2\left(\frac{\sigma}{r_{ij}}\right)^{13} - \left(\frac{\sigma}{r_{ij}}\right)^{7}\right](\boldsymbol{q}_i - \boldsymbol{q}_j). \end{aligned}$$

Now introduce the change of variables

$$\boldsymbol{Q}_i = \sigma^{-1}\boldsymbol{q}_i, \qquad i = 1, 2, \ldots, \mathrm{N},$$

then $\dot{\boldsymbol{q}}_i = \sigma\dot{\boldsymbol{Q}}_i$, $\ddot{\boldsymbol{q}}_i = \sigma\ddot{\boldsymbol{Q}}_i$, and $\|\boldsymbol{q}_i - \boldsymbol{q}_j\| = \sigma\|\boldsymbol{Q}_i - \boldsymbol{Q}_j\|$. The equations of motion in these normalized coordinates may be written

$$\frac{m\sigma^2}{\epsilon}\frac{d^2\boldsymbol{Q}_i}{dt^2} = -\sum_{\substack{j=1\\ j\neq i}}^{\mathrm{N}} \frac{\hat{\varphi}'_{\mathrm{LJ}}(R_{ij})}{R_{ij}}(\boldsymbol{Q}_i - \boldsymbol{Q}_j),$$

where $R_{ij} = \|\boldsymbol{Q}_i - \boldsymbol{Q}_j\|$ and $\hat{\varphi}_{\mathrm{LJ}}(R) = 4[R^{-12} - R^{-6}]$. Thus we see that if we make the additional time transformation

$$\tau = \alpha t, \qquad \alpha^2 = \frac{\epsilon}{m\sigma^2},$$